

Muhammad Kashif

PhD Researcher / Multimodal Healthcare AI, Physiological Time-Series, Wearable Sensing and Digital Health

Rende (CS), Italy
✉ kashif.muhammad@dimes.unical.it
in kashifhon
🔗 Google Scholar

Research Profile

AI-for-healthcare researcher focused on multimodal physiological and wearable data, continuous glucose monitoring, speech-based neurological assessment, and longitudinal health time-series analysis. Experienced in Python/PyTorch-based machine- and deep-learning workflows, subject-independent validation, event-level evaluation, and multimodal sensor analysis. Research interests include digital health, precision medicine, physiological signal modelling, multimodal healthcare AI, and clinically meaningful validation of AI systems.

Research Experience

- 10/2024–04/2025 **International Scholar**, *KU Leuven, e-Media Research Lab, ESAT-STADIUS @ Groep T*, Leuven, Belgium
Worked on AI-based healthcare monitoring using CGM, IMU, and wearable/physiological time-series data under the supervision of Prof. Bart Vanrumste and Dr. Chunzhuo Wang. Developed multimodal machine-learning pipelines for wearable sensor analysis, including IMU-based eating-event detection on the Clemson All-Day Dataset, temporal alignment, event annotation, preprocessing, and subject-level evaluation.
- 2023–2026 **PhD Researcher**, *University of Calabria, DIMES*, Rende, Italy
Conducted doctoral research on AI methods for multimodal healthcare data, including continuous glucose monitoring, glucose prediction, meal-event detection, insulin-related temporal modelling, wearable sensing, and speech-based neurological assessment. Developed LSTM/BiLSTM and deep-learning pipelines with preprocessing, windowing, feature engineering, event decoding, subject-independent validation, and clinically relevant evaluation.
- 06/2020–05/2023 **Research Assistant**, *Artificial Intelligence & Data Analytics Lab, Prince Sultan University*, Riyadh, Saudi Arabia, Remote
Worked on AI-based healthcare and biomedical data analysis, including facial-expression analysis, COVID-19 prediction, lung-cancer detection, breast-cancer diagnosis, and IoT-based healthcare applications.

Selected Research Contributions

Multimodal physiological and wearable AI: Developed machine- and deep-learning pipelines for CGM, insulin, IMU, wearable, and speech data, with emphasis on temporal modelling, multimodal integration, and subject-independent evaluation.

CGM-based prediction and event detection: Developed LSTM/BiLSTM frameworks for glucose forecasting and meal-episode detection using CGM and insulin-related features, including constraint-based decoding and event-level evaluation.

Speech-based neurological assessment: Developed deep-learning approaches for neurological assessment from speech, including LSTM-based multiple-sclerosis classification and task-conditioned WavLM/LoRA with prompt-aligned ASR error fusion for Parkinsons assessment.

Education

- 2023–2026 **Ph.D. in Information and Communications Technology**, *University of Calabria, DIMES*, Rende, Italy
Doctoral programme completed; thesis defence pending. Expected PhD degree before 15 November 2026.
Thesis: AI Methods for Physiological Time-Series Analysis in Healthcare: CGM Analytics and Speech-Based Neurological Assessment
Supervisors: Prof. Sergio Flesca, Prof. Pierangelo Veltri
- 2016–2019 **M.S. in Computer Science**, *International Islamic University Islamabad*, Islamabad, Pakistan
Thesis: Facial Expression Recognition Using Patch-Based Multiple LBP Descriptors in Unconstrained Environments

Selected Publications

- M. Kashif**, P. Vizza, G. Tradigo, and S. Flesca, Task-conditioned PEFT with prompt-aligned ASR error structure fusion for Parkinson's speech assessment, in *Proceedings of the 17th ACM International Conference on Bioinformatics, Computational Biology and Health Informatics*, 2026, pp. 1-2.
- M. Kashif**, C. Wang, S. Flesca, and P. Veltri. Meal Episode Detection from Continuous Glucose Monitoring Using a BiLSTM Sliding-Window and Constraint-Based Decoding. *34th European Signal Processing Conference (EUSIPCO)*, 2026, pp. 856-860.
- M. Kashif**, C. Wang, S. Flesca, and P. Veltri. AI-Based Detection and Prediction of Eating Events Using Continuous Glucose Monitoring: A Comprehensive Review. *Biomed. Signal Process. Control*, vol. 127, article no. 111207, 2026.
- M. Kashif** et al., Using LSTM-based model on vocal signal analysis for the classification of multiple sclerosis, in *Lecture Notes in Computer Science*, Cham: Springer Nature Switzerland, 2025, pp. 192-196.
- M. Kashif**, S. Flesca, and P. Veltri. GLSTM: Using LSTM for Glucose Level Prediction. In *pHealth 2024: Proceedings of the 20th International Conference on Wearable Micro and Nano Technologies for Personalized Health - Rende, Italy*, IOS Press, 2024, pp. 103-107.

Technical Skills

Programming & ML: Python, PyTorch, TensorFlow/Keras, scikit-learn, NumPy, Pandas, SciPy, Git, Linux **Healthcare AI:** physiological time-series modelling, multimodal learning, CGM analytics, wearable/IMU sensing, speech-based health AI, signal preprocessing, event detection, digital biomarkers **Methods:** LSTM/BiLSTM/GRU, Conv1D, WavLM, Whisper, LoRA/PEFT, feature engineering, statistical modelling, subject-independent validation, event-level evaluation, reproducible ML pipelines

Selected Training and Awards

- 2026 **International Summer School on Wearable Sensors**, *Università Campus Bio-Medico di Roma*, Rome, Italy
- 2026 **ELLIS Winter School on Foundation Models**, *University of Amsterdam*, Amsterdam, The Netherlands
- 2023–2025 **Data Science Summer Schools**, *Hertie School*, Berlin, Germany
Data science, statistical learning, machine learning, and applied computational methods.
- 2026 **Best Poster Award**, *ACM BCB*, University of Calabria, Italy
- 2023–2026 **Fully Funded PhD Scholarship**, *FAIR – Future Artificial Intelligence Research Project, PNRR*, University of Calabria, Italy